Domination in Planar Graphs with Small Diameter

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Dedicated to Professor Henda Swart, on her 60th birthday

Abstract

MacGillivray and Seyffarth (J. Graph Theory **22** (1996), 213–229) proved that planar graphs of diameter two have domination number at most three and planar graphs of diameter three have domination number at most ten. They also give examples of planar graphs of diameter four having arbitrarily large domination numbers. In this paper we improve on their results. We prove that there is in fact a unique planar graph of diameter two with domination number three, and all other planar graphs of diameter two have domination number at most two. We also prove that every planar graph of diameter three and of radius two has domination number at most six. We then show that every sufficiently large planar graph of diameter three has domination number at most seven. Analogous results for other surfaces are discussed.

1 Introduction

Domination and its variations in graphs are now well studied (see [1, 4, 5]). However, the original domination number of a graph continues to attract attention. Many bounds have been proven and results obtained for special classes of graphs such as cubic graphs and products of graphs. On the other hand, the decision problem to determine the domination number of a graph remains NP-hard even when restricted to cubic graphs or planar graphs of maximum degree 3 [3]. Hence it is of interest to determine upper bounds on the domination number of a graph. Bounds on the domination number of a graph in terms of its order and

*Research supported in part by the University of Natal and the National Research Foundation.

minimum degree can be found, for example, in [8, 9, 11] and bounds on the domination number of a graph in terms of its size can be found in [12] and elsewhere.

In this paper we consider the domination of planar graphs with small diameter. It is trivial that a tree of radius 2 and diameter 4 can have arbitrarily large domination number. So the interesting question is what happens when the diameter is 2 or 3. As pointed out in [7], the restriction of bounding the diameter on the domination of a planar graph is reasonable to impose because planar graphs with small diameter are often important in applications (see [2]). MacGillivray and Seyffarth [7] proved that planar graphs with diameter two or three have bounded domination numbers. In particular, this implies that the domination number of such a graph can be determined in polynomial time. On the other hand, they observed that in general graphs with diameter 2 have unbounded domination number.

In this paper we improve on the results of [7]. We show that there is a unique planar graph of diameter 2 with domination number 3. Hence every planar graph of diameter 2, different from this unique planar graph, has domination number at most 2. We then prove that every planar graph of diameter 3 and of radius 2 has domination number at most 6. We then show that every sufficiently large planar graph of diameter 3 has domination number at most 7.

For notation and graph theory terminology we in general follow [1]. So, for a graph G, if $X, Y \subseteq V(G)$, then we say that X *dominates* Y if every vertex of $Y - X$ is adjacent to some vertex of X. In particular, if X dominates $V(G)$, then X is called a *dominating set* of G. The *domination number* of G, denoted by $\gamma(G)$, is the minimum cardinality of a dominating set. If a vertex u is adjacent with a vertex v, then to simplify the notation, we write $u \sim v$, while if u and v are nonadjacent, we write $u \not\sim v$. A path is a sequence of pairwise adjacent vertices with no repeated vertex.

2 Main Results

MacGillivray and Seyffarth established the following result.

Theorem 1 *[7] The domination number of a planar graph of diameter* 2 *is bounded above by* 3.

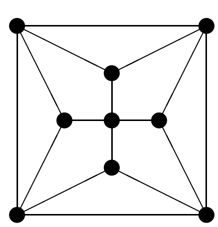

Figure 1: The planar graph F of diameter 2 with domination number 3

The bound of Theorem 1 is sharp as may be seen by considering the graph F of Figure 1 constructed by MacGillivray and Seyffarth [7].

Our aim is to characterize planar graphs of diameter two with domination number 3. We shall prove:

Theorem 2 *Every planar graph of diameter* 2 *has domination number at most* 2 *except for the graph* F *of Figure 1 which has domination number* 3.

That is, the graph of Figure 1 is the unique planar graph of diameter two with domination number 3. We prove Theorem 2 in Section 3. In brief, we first establish the existence of a 4-cycle. Then we show that this cycle is not both induced and dominating, nor both non-induced and dominating, and therefore not dominating. Finally, we show that it follows that G is isomorphic to F.

MacGillivray and Seyffarth [7] also proved that planar graphs with diameter 3 have bounded domination numbers.

Theorem 3 *[7] The domination number of a planar graph of diameter* 3 *is bounded above by* 10.

It is not known in [7] whether the bound given in Theorem 3 is sharp. MacGillivray and Seyffarth [7] give an example of a planar graph of diameter 3 with domination number 6. In fact, their example can be used to construct planar graph of diameter 3 of arbitrarily large order with domination number 6. See Figure 2.

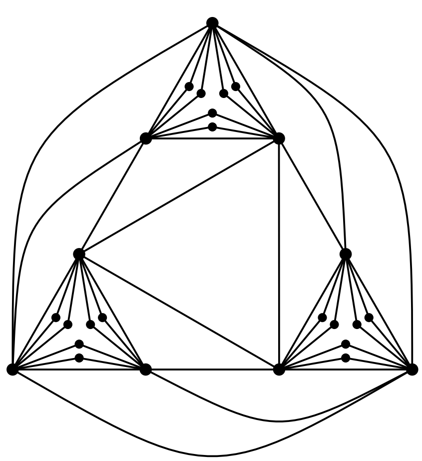

Figure 2: A planar graph with diameter 3 and domination number 6

Our aim is to improve the bound in Theorem 3. We shall prove:

Theorem 4 *Every planar graph of diameter* 3 *and of radius* 2 *has domination number at most* 6.

Theorem 5 *Every sufficiently large planar graph of diameter* 3 *has domination number at most* 7*.*

We prove Theorem 4 in Section 4 and Theorem 5 in Section 5. In brief, to prove Theorem 4, we define basic cycles as cut-cycles with certian special properties. We then show the existence of a basic cycle of length at most 5. The proof is completed by proving a series of lemmas to bound the domination number when there exists a short basic cycle. To prove Theorem 5, we show that for a sufficiently large planar graph G of radius and diameter 3, there exists a planar graph G' of radius at most 2 and diameter at most 3 such that $\gamma(G) \leq \gamma(G') + 1$. Theorem 5 then follows by applying Theorem 4 to the graph G'.

In Section 6 it is shown that these results are typical of other surfaces. In particular, for each orientable surface there are only finitely many graphs with diameter 2 and domination number more than 2. Also, for each orientable surface the domination number of graphs of diameter 3 is bounded.

3 Proof of Theorem 2

To prove Theorem 2, suppose G is a planar graph of diameter two satisfying $\gamma(G) > 2$. If a and b are two vertices in G, then there is always a vertex not dominated by $\{a, b\}$. We shall denote one such vertex by v_{ab}.

Fix an embedding G^* of G in the plane. From the Jordan Closed Curve Theorem, we know that a cycle C in G^* separates the plane into two regions, which we call the *sides* of C. Vertices of different sides of C are said to be separated by C. The side of C that consists of the unbounded region we call the *outside of* C, while the side of C that consists of the bounded region we call the *inside of* C. If C has length n and there are vertices both inside and outside C, then we say that C is a *cut-n-cycle*. A cut-3-cycle is also called a *cut-triangle*. Since a cut-set dominates a graph of diameter 2, it follows that G is 3-connected; therefore G has an essentially unique embedding in the plane and so we may speak of cut-cycles of G rather than of G^*.

The theorem is established by a series of lemmas. The first lemma establishes the existence of a 4-cycle. Then we show that this cycle is not both induced and dominating (Lemma 3.5), nor both non-induced and dominating (Lemma 3.8), and therefore not dominating. Finally, we show that it follows that G is isomorphic to F (Lemma 3.9).

Lemma 3.1 *There is at least one* 4*-cycle in* G*.*

Proof. Since G is 3-connected, G has minimum degree at least 3. In particular, G has a cycle. Since the diameter of G is 2, the girth of G is at most 5. If the girth of G is 5, then by [10] G must be a Moore graph. By [6], the only planar Moore graph of diameter 2 is the 5-cycle—which has domination number 2. If the girth is 4 then we are done.

So assume that G has girth 3 and a triangle $C\colon a, b, c, a$. Then the neighbourhoods of a, b and c outside C are disjoint (else we immediately have a 4-cycle). Let a' and b' be neighbours not on C of a and b, respectively. Then vertices a' and b' are not adjacent, else we immediately have a 4-cycle.

Since $d(a', b') = 2$, vertices a' and b' have a common neighbour, m say. Then vertex m is not adjacent to c, else again we have a 4-cycle. Since $d(c, m) = 2$, there is a new vertex c' adjacent to both c and m. Consider a vertex v other than $\{a, b, c\}$ not dominated by m. It lies outside the triangle a, b, c, a (to reach m); without loss of generality, v is inside the 5-cycle a, b, b', m, a', a. But then v and c' must have a common neighbour. Hence there is an edge from c' to a' or from c' to b' and thus a 4-cycle. □

Lemma 3.2 *If C is a cut-triangle of G, then one side of C contains exactly one vertex and this vertex is adjacent to all three vertices of C.*

Proof. Suppose $C\colon a, b, c, a$ is a cut-triangle of G. Without loss of generality, the vertex v_{ab} not dominated by $\{a, b\}$ lies outside C. Then every vertex inside C must be adjacent to c, since G has diameter 2. Hence vertices v_{ac} and v_{bc} both lie outside C. Therefore, every vertex inside C is adjacent to all of a, b, c. Since G is planar, there is therefore exactly one vertex inside C. □

Lemma 3.3 *If C is a cut-4-cycle of G, then one side of C contains exactly one vertex and this vertex is adjacent to at least three vertices of C.*

Proof. Suppose $C\colon a, b, c, d, a$ is a cut-4-cycle of G. Since G has diameter 2, vertices v_{ac} and v_{bd} must lie on the same side of C, without loss of generality outside C. By planarity it cannot hold that v_{ac} is adjacent to both b, d and v_{bd} is adjacent to both a, c. Without loss of generality, $v_{bd} \not\sim c$. It follows that every vertex inside C is adjacent to a and to at least one of b, d. Therefore v_{ab} and v_{ad} must lie outside C. Consequently, every vertex inside C is adjacent to at least one of c, d and to at least one of b, c. Thus, every vertex inside C is adjacent to a and to at least two of b, c, d.

It follows that there can be at most two vertices inside C. Moreover, if there are two vertices inside C, then one is adjacent to a, b, c and the other to a, c, d. We have therefore shown that for any cut-4-cycle, on the one side there are at most two vertices, and each is adjacent to at least three vertices of the cut-4-cycle.

Now suppose there are two vertices inside C. Then v_{ac} must be adjacent to both b and d. Now, consider the 4-cycle $D\colon a, b, v_{ac}, d, a$. Without loss of generality vertex c lies inside D. Then the inside of D contains c and at least two other vertices, so the outside of D has at most two vertices and each of these is adjacent to at least three vertices of D. However, v_{bd} belongs to the outside of D and is adjacent to at most two vertices of D, a contradiction. Hence C must contain exactly one vertex. □

Lemma 3.4 *There is no dominating induced 4-cycle in G.*

Proof. Suppose $C\colon a, b, c, d, a$ is a dominating induced 4-cycle of G. As in the proof of Lemma 3.3, we may assume v_{ac} and v_{bd} both lie outside C and a is the only neighbour of

v_{bd} on C. Since $d(v_{bd}, c) = 2$, there exists a new vertex e adjacent to v_{bd} and c. Without loss of generality d is outside the cycle a, v_{bd}, e, c, b, a in G^*, and also, without loss of generality, v_{ac} is inside that cycle. Since $d(v_{ac}, d) = 2$, e must be adjacent to v_{ac} and d.

Now, consider the vertex v_{ae}. By Lemma 3.2, v_{ae} cannot lie inside the triangle c, d, e, c. By Lemma 3.3, v_{ae} cannot lie outside the 4-cycle a, d, e, v_{bd}, a. Since every vertex inside C is adjacent to both a and b, v_{ae} cannot lie inside C. Furthermore, v_{ae} cannot lie inside the 5-cycle $a, v_{bd}, e, v_{ac}, b, a$, for otherwise $d(v_{ae}, d) > 2$. Hence v_{ae} must lie inside the 4-cycle b, c, e, v_{ac}, b. Similarly, v_{be} must lie outside the 4-cycle a, d, e, v_{bd}, a. But then $d(v_{ae}, v_{be}) > 2$, a contradiction. Hence G contains no dominating induced 4-cycle. □

Lemma 3.5 *There is no induced cut-4-cycle in G.*

Proof. Suppose C is an induced cut-4-cycle in G. By Lemma 3.4, there is a vertex x not adjacent to any vertex of C. But then x is at distance at least 3 from a vertex that is separated from x by C, a contradiction. □

Lemma 3.6 *There is no cut-triangle in G.*

Proof. Suppose $C\colon a, b, c, a$ is a cut-triangle in G. By Lemma 3.2, we may assume the inside of C contains one vertex x, say, which is adjacent to all three vertices of C. The vertices that lie outside C we shall call exterior vertices. Thus every exterior vertex has at least one neighbour on C.

Let A', B' and C' denote the set of exterior vertices adjacent only to a, b and c, respectively, on C. Since no pair of vertices of C dominates, all three of A', B' and C' are nonempty. Let a', b' and c' be members of each set, respectively.

Suppose there exists an exterior vertex y adjacent to all three vertices of C. Then $\{a, b, c, x, y\}$ induces a triangulation of the plane in which each triangle contains two vertices of C. So the region containing a' is bounded by such a cut-triangle and a' is therefore adjacent to two vertices of C, a contradiction. Hence, each exterior vertex is adjacent to at most two vertices of C.

Consider the edges (if any) between A' and B'. Suppose there are two such edges a_1b_1 and a_2b_2 where $a_1, a_2 \in A'$, $b_1, b_2 \in B'$, and possibly $a_1 = a_2$ or $b_1 = b_2$. Then both the 4-cycles a, b, b_1, a_1, a and a, b, b_2, a_2, a are induced, and at least one is a cut-cycle. This contradicts Lemma 3.5. Hence, there is at most one edge between A' and B'. Similarly, there is at most one edge between A' and C', and between B' and C'.

Suppose w_{ab} and w'_{ab} are both exterior vertices adjacent to both a and b. Then for at least one of the 4-cycles a, c, b, w_{ab}, a and a, c, b, w'_{ab}, a, there are at least two vertices both inside and outside the cycle, contrary to Lemma 3.3. So there is at most one exterior vertex adjacent to both a and b. Similarly, there is at most one exterior vertex adjacent to both a and c (respectively, to b and c). If there exists an exterior vertex adjacent to both a and b, then we call it w_{ab}, and similarly with w_{ac} and w_{bc}.

We now consider some properties of w_{ab}. See Figure 3. First, by Lemma 3.3 applied to cycle a, c, b, w_{ab}, a, the inside of triangle a, b, w_{ab}, a is empty. Second, by Lemma 3.5, there is no edge between A' and B'. Third, since both a, w_{ab}, c', c, a and a, w_{ab}, w_{bc}, c, a would be induced cut-4-cycles, Lemma 3.5 implies that the neighbours of w_{ab} not on C are restricted to $A' \cup B'$. In particular, the path of length 2 from c' to w_{ab} is via $A' \cup B'$. Similar statements hold for w_{ac} and w_{bc}, if they exist.

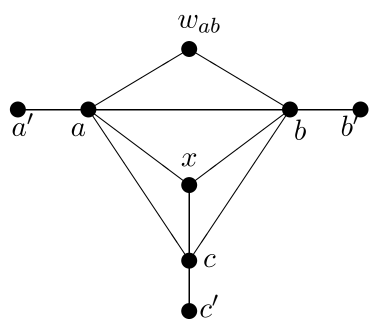

Figure 3: A subgraph of G

Let W be the set of exterior vertices adjacent to two vertices of C. If W is not empty, then W is an independent set of cardinality at most 3. We now consider a case study on how many vertices there are in W.

Suppose $|W| = 3$. Then there is no edge between any two of the sets A', B', C'. Hence, since W is independent, $d(c', w_{ab}) > 2$ for example, a contradiction.

Suppose $|W| = 2$. We may assume $W = \{w_{ab}, w_{ac}\}$. Then there is no edge between A' and B', and no edge between A' and C'. Then a shortest path from b' to w_{ac} must be via a vertex of C', say c'. Thus $b'c'$ is the only edge between B' and C'. A shortest path from c' to w_{ab} must be via b'. By Lemma 3.5, the inside of the induced 4-cycle b, c, c', b', b is empty. Now, the pair $\{a, c'\}$ does not dominate, and so there is a vertex e adjacent to neither a nor c'. Necessarily, $e \in B' \cup C'$. By Lemma 3.2, e cannot lie inside the triangles c, c', w_{ac}, c or a, c, w_{ac}, a (for otherwise e would be adjacent to c' or a, respectively). Hence e cannot be adjacent to c and therefore must lie inside the triangle b, b', w_{ab}, b. Similarly, the vertex f not dominated by the pair $\{a, b'\}$ is inside the triangle c, c', w_{ac}, c. But then $d(e, f) > 2$, a contradiction.

Suppose $|W| = 1$. We may assume $W = \{w_{ab}\}$. Then there is no edge between A' and B'. Also, a shortest path from c' to w_{ab} must be via a vertex of A' or B'. We may assume a' is adjacent to c' and w_{ab}. By Lemma 3.5, the inside of the induced 4-cycle a', a, c, c', a' is empty. Now, there is a vertex adjacent to neither c nor w_{ab}: it must lie in the side of c, c', a', w_{ab}, b, c not containing a and hence is a member of B', say b'. Since $d(a', b') = 2$, it holds that $b' \sim c'$. Since the inside of the induced 4-cycle b', b, c, c', b' is empty, C' consists only of the vertex c'. But then the pair $\{a', b\}$ dominates, a contradiction. (Any other member of A' is inside the cycle a, a', w_{ab}, a and hence adjacent to a'.)

Suppose W is empty. Then there must be an edge between at least two of the sets A', B' and C', for otherwise a' would be at distance at least 3 from b' or c'. We may assume $a'b'$ and $b'c'$ are edges. Then $a'b'$ is the only edge between A' and B', and $b'c'$ is the only edge between B' and C'. Furthermore, the inside of the induced 4-cycles a, b, b', a', a and b, c, c', b', b are empty. Thus b' is the unique vertex of B. For the pair $\{a, c'\}$ not to dominate, there must be a vertex $c'' \in C'$ different from c'. Since $d(b', c'') = 2$, a shortest path from b' to c'' must be via a'. Since the inside of the induced 4-cycle a, c, c'', a', a is empty, a' is the unique vertex of A'. But now $\{a', c\}$ dominates G, a contradiction.

Since all four possibilities for $|W|$ produce a contradiction, we deduce that there can be no cut-triangle in G. $\square$

Lemma 3.7 *There is no cut-4-cycle in G.*

Proof. Suppose $C\colon a, b, c, d, a$ is a cut-4-cycle in G. By Lemma 3.5, C cannot be an induced 4-cycle. We may assume that ac is an edge. But then either a, b, c, a or a, c, d, a is a cut-triangle in G, contradicting Lemma 3.6. $\square$

Lemma 3.8 *There is no dominating non-induced 4-cycle in G.*

Proof. Suppose $C\colon a, b, c, d, a$ is a dominating 4-cycle of G with $a \sim c$. We may assume that the edge ac lies inside C in G^*. Then, by Lemma 3.6, the inside of C is empty. If $b \sim d$, then by Lemma 3.6, the outside of C is empty too, a contradiction. Hence we may assume that $b \not\sim d$. By Lemma 3.7, there is no vertex not on C adjacent to both a and c, nor to both b and d. Let A', B', C' and D', respectively, denote the set of vertices whose only neighbour on the cycle C is a, b, c and d.

We consider two possibilities.

Case 1: There is an edge between A' and B'.

Suppose $a' \sim b'$ where $a' \in A'$ and $b' \in B'$. Since $d(b', d) = 2$, there exists a new vertex e adjacent to b' and d. By the lack of cut-4-cycles, $e \not\sim a$ and $e \not\sim b$. See Figure 4.

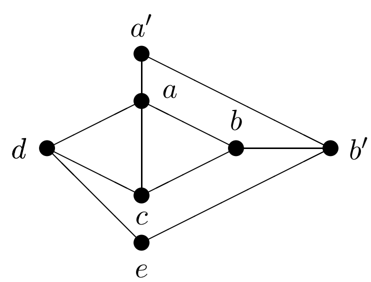

Figure 4: A possible subgraph of G^*

By Lemma 3.4, there exists a vertex f not dominated by the 4-cycle a, a', b', b, a. If f lies outside the cycle a, a', b', e, d, a, then $d(b, f) > 2$. So f must lie inside the cycle b, b', e, d, c, b. Now a shortest a'–f path must be via e. Also, $e \not\sim c$.

If $c \sim f$, then v_{ce} can only lie inside the cycle b, b', e, f, c, b and a shortest d–v_{ce} path must be via f. If $c \not\sim f$, then d is adjacent to f since C dominates. In either case, $d \sim f$.

The vertex v_{ce} can only lie inside the cycle e, b', b, c, d, f, e. A shortest a'–v_{ce} path must be via b'. By the lack of cut-4-cycles, $d \not\sim v_{ce}$. Since the vertices of C dominate, $b \sim v_{ce}$.

Now, v_{be} must lie inside the cycle $b, v_{ce}, b', e, f, d, c, b$. A shortest a'–v_{be} path must be via b'. But then, by lack of cut-4-cycles, $c \not\sim v_{be}$ and $d \not\sim v_{be}$, whence $d(a, v_{be}) > 2$, a contradiction.

Case 2: There is no edge between any two sets in $\{A', B', C', D'\}$.

Since $\{b, d\}$ does not dominate G, one of A' or C' is nonempty; say $a' \in A'$. Since $\{a, c\}$ does not dominate, one of B' or D' is nonempty; say $b' \in B'$. Since $d(a', b') = 2$, there exists a new vertex f adjacent to a' and b'. As we are not in Case 1, the vertex f is not in $A' \cup B' \cup C' \cup D'$ and thus has at least two neighbours on the cycle C. The vertex c is not a neighbour of f by lack of cut-4-cycles. By lack of cut-4-cycles, f cannot have both a and d as neighbours nor both b and d as neighbours. Hence a and b are the only neighbours of f on C.

Let e be a common neighbour of b' and d. See Figure 5. Suppose $a' \sim e$ and assume b lies outside the 4-cycle a, d, e, a', a in G^*. By Lemma 3.7, v_{ae} cannot lie inside the 4-cycle a, d, e, a', a, nor outside the 4-cycle a', f, b', e, a'. So v_{ae} lies inside the cycle c, b, b', e, d, c, and $d(v_{ae}, a') > 2$, a contradiction. Hence $a' \not\sim e$.

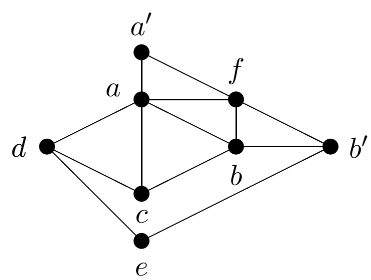

Figure 5: A possible subgraph of G^*

Suppose $e \sim f$. Assume b lies outside the cycle e, f, b', e in G^*. Then v_{ae} cannot be inside the cycle b, c, d, e, b', b, for otherwise it would be at distance at least 3 from a'. So v_{ae} lies outside the cycle a, d, e, f, a', a. Then $d \sim v_{ae}$, and since $d(v_{ae}, b') = 2$, $f \sim v_{ae}$. But that produces a cut-4-cycle, a contradiction. Hence $e \not\sim f$.

Since $d(a', e) = 2$, there exists a new vertex g adjacent to both a' and e. But then if v_{ae} lies inside a, a', g, e, d, a, then $d(v_{ae}, b) > 2$, while if v_{ae} lies inside b, b', e, d, c, b, then $d(v_{ae}, a') > 2$. This produces a contradiction. $\square$

Lemma 3.9 *If there is a nondominating* 4*-cycle in* G*, then* G *is the graph* F *of Figure 1.*

Proof. Suppose $C\colon a, b, c, d, a$ is a 4-cycle in G and vertex m is not dominated by C. We draw G^* with m outside C. By Lemma 3.6, it cannot hold that both bd and ac are edges. Without loss of generality $b \not\sim d$.

Let W denote the set of vertices outside C with two neighbours on the cycle C: these two neighbours must be consecutive by lack of cut-4-cycles (G cannot contain $K(2, 3)$). By the lack of cut-triangles, if the vertex w_{ab} adjacent to a and b exists then it is unique, and similarly with w_{bc}, w_{cd} and w_{ad}.

We consider three cases.

Case 1: At most one neighbour of m is in W.

If no neighbour of m is in W, then the four paths of length 2 from m to C are internally disjoint and the middle vertex on the m–b path is too far from d. So we may assume that at least one neighbour of m is in W: say w_{ab} adjacent to a and b.

Let c' and d' denote the interior vertices on c–m and d–m shortest paths, respectively. Consider the shortest d'–b path: since only one neighbour of m is in W, this path is via w_{ab} so that $d' \sim w_{ab}$. See Figure 6.

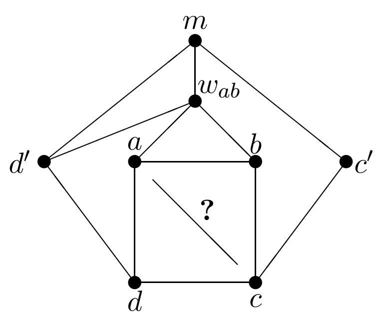

Figure 6: m with one neighbour in W

Consider any other vertex x. By lack of cut-triangles and cut-4-cycles, x cannot lie inside the cycle $m, w_{ab}, b, c, d, d', m$. If x lies inside the cycle m, c', c, b, w_{ab}, m, then since $d(x, d) = 2$, x must be adjacent to c. If x lies outside the cycle m, c', c, d, d', m, then since $d(x, b) = 2$, x must be adjacent to c. Either way, $x \sim c$. Hence $\{c, w_{ab}\}$ dominates G, a contradiction.

Case 2: Two neighbours of m are in W but opposite; say w_{ab} and w_{cd}.

Then, $w_{ab} \not\sim w_{cd}$, by lack of cut-4-cycles. Let f be a vertex not dominated by $\{a, w_{cd}\}$ and g a vertex not dominated by $\{c, w_{ab}\}$. (We shall show them to be distinct.) Suppose f lies outside the cycle $a, d, w_{cd}, m, w_{ab}, a$. Then a shortest b–f path must be via w_{ab}, and a shortest c–f path must be via d. In particular, f is adjacent to both w_{ab} and d, while $f \neq g$. If g lies outside the cycle $w_{ab}, m, w_{cd}, d, f, w_{ab}$, then $d(b, g) > 2$. Hence g must lie inside the cycle $b, c, w_{cd}, m, w_{ab}, b$. A shortest a–g path must be via b, a shortest d–g path must be via w_{cd}, while a shortest f–g path must be via m. Thus every region of the graph induced by the nine vertices identified so far is a triangle or a 4-cycle, and therefore there cannot be additional vertices in G^*. But then $\{w_{ab}, w_{cd}\}$ dominates G, a contradiction. Hence f must lie inside the cycle $w_{ab}, b, c, w_{cd}, m, w_{ab}$. Similarly, g must lie outside the cycle $w_{ab}, a, d, w_{cd}, m, w_{ab}$. See Figure 7.

A shortest f–d path must be via c, a shortest g–b path via a, and a shortest f–g path via m. Now let x be any other vertex. Without loss of generality, x lies inside the cycle c, f, m, w_{ab}, b, c. Then x is adjacent to m (to reach g) and adjacent to c (to reach d). But that means x, c, w_{cd}, m, x is a cut-4-cycle, a contradiction. So there is no other vertex and G^* has nine vertices. Since $\{c, m\}$ does not dominate, it holds that $a \not\sim c$. Thus $f \sim b$ and

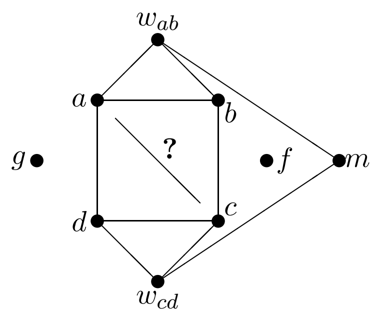

Figure 7: m with opposite neighbours in W

$g \sim d$. That is, F is a spanning subgraph of G. It is easily checked that adding any edge to F decreases the domination number. So, in fact $G \cong F$, as required.

Case 3: Two neighbours of m are in W but not opposite.

Without loss of generality, w_{ab} is one of them.

Suppose $a \sim c$ and m's other neighbour is w_{ad}. Let c' denote the interior vertex on a c–m shortest path. Since $c \not\sim w_{ab}$ and $c \not\sim w_{ad}$, c' is distinct from w_{ab} and w_{ad}. Thus the vertex not dominated by $\{c, m\}$ is too far from either b or d, a contradiction. So we may assume m's other neighbour is w_{bc} (if $a \not\sim c$ then the two possibilities are symmetric). Let d' be the common neighbour of m and d. Since $d(b, d') = 2$, we may assume that $d' \sim w_{ab}$. See Figure 8.

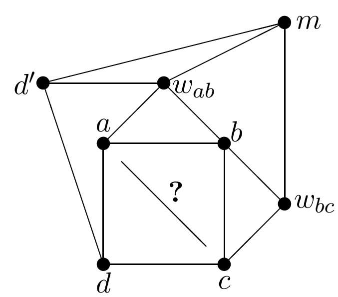

Figure 8: m with non-opposite neighbours in W

Suppose $a \sim c$. Then consider a vertex x not dominated by $\{c, d'\}$. This must lie outside the cycle d, d', m, w_{bc}, c, d and must be adjacent to d (to reach a), to w_{bc} (to reach b) and to m (to reach w_{ab} since by lack of cut-4-cycles, $w_{ab} \not\sim w_{bc}$). But then G^* has nine vertices, and the pair $\{c, m\}$ dominates, a contradiction. Hence $a \not\sim c$. In particular, since this is the final case, we may assume every 4-cycle is induced, and thus that $w_{ab} \not\sim w_{bc}$.

Now, consider a vertex d'' not dominated by $\{b, d'\}$. The vertex d'' must lie outside the cycle d, d', m, w_{bc}, c, d, and hence be adjacent to d (to reach a), to m (to reach w_{ab}), and to w_{bc} (to reach b as we're not in Case 2). Thus F is a subgraph of G, and as in Case 2, $G \cong F$, as required. $\square$

This completes the proof of Theorem 2.

4 Proof of Theorem 4

In this section we prove Theorem 4. As in the proof of Theorem 2, we consider cut-cycles. Their importance stems from the fact that in a planar graph of diameter 3, there cannot on both sides of a cut-cycle be vertices not dominated by the cycle.

In particular, we define a *basic cycle* as follows. Let vertex x have eccentricity 2 in G. Then a basic cycle C is an induced cycle $x, v_1, v_2, \ldots, v_r, x$ such that on both sides of the cycle there is a vertex whose neighbours on the cycle are a subset of the two consecutive vertices fartherest from x, specifically $v_{(r-1)/2}$ and $v_{(r+1)/2}$ if r is odd, and $v_{r/2}$ and $v_{r/2+1}$ if r is even.

Our strategy is as follows. In Subsection 4.1 we show the existence of a basic cycle of length at most 5. Thereafter, in Subsection 4.2 we prove a series of lemmas, while in Subsection 4.3 we complete our proof of Theorem 4 by using the crucial Lemma 4.4 to bound the domination number when there exists a short basic cycle.

4.1 Basic Cycles Exist

Lemma 4.1 *Let G be a plane graph of radius 2 and diameter 3 with central vertex x. Then $\gamma(G) \leq 6$ or there exists a basic cycle of length at most 5.*

Proof. Suppose that there is no basic cycle of length at most 5 in G. We show then that $\gamma(G) \leq 6$; that is, there is a set of six vertices that dominates G.

Let $Y = V(G) - N[x]$. Let M be a minimal subset of $N(x)$ that dominates Y. The set M exists since x has eccentricity 2. Let $|M| = m$. Since $\gamma(G) \leq m+1$, we may assume $m \geq 6$.

Let the vertices of M be $n_0, n_1, \ldots, n_{m-1}$ in cyclic order (clockwise) around x in G. Let Y_i' be the set of vertices of Y whose only neighbour in M is n_i. By the minimality of M, each Y_i' is nonempty. Let $Y_0, Y_1, \ldots, Y_{m-1}$ be a partition of Y such that $Y_i \subseteq N(n_i)$ for each i. Necessarily, $Y_i' \subseteq Y_i$ for each i. If there is a vertex of Y_i' adjacent to both a vertex of Y_{i-1} and a vertex of Y_{i+1} (where addition is taken modulo m), then this vertex is unique by the planarity of G and is denoted by y_i. Otherwise, we let y_i be any vertex of Y_i'.

We say that two neighbours u_1 and u_2 of x are *separated* if there is a vertex of M between u_1 and u_2 in both directions around x in the embedding of G. We define type-1, type-2 and type-3 edges as follows. A *type*-1 edge joins vertices $u_1, u_2 \in N(x)$ such that u_1 and u_2 are separated. A *type*-2 edge joins vertices $u_1 \in N(x)$ and $v_2 \in Y$ with v_2 dominated by a vertex u_2 of M such that u_1 and u_2 are separated. A *type*-3 edge joins vertices $v_1, v_2 \in Y$ with v_1 and v_2 dominated by vertices u_1 and u_2 of M respectively such that u_1 and u_2 are separated.

Claim 4.1.1 *(i) There is no type-1, type-2 or type-3 edge.*
(ii) If there is an edge from Y_i to Y_j, then i and j are consecutive.

Proof. (i) Let e be an edge. If $e = u_1u_2$ is a type-1 edge, then there is a vertex n_i of M both inside and outside the cycle $C : x, u_1, u_2, x$. Since the vertex y_i is not dominated by $\{x\}$, C is a basic triangle, a contradiction.

If $e = u_1v_2$ is a type-2 edge, then again there is a vertex y_i both inside and outside the cycle $C : x, u_1, v_2, u_2, x$ not dominated by $\{x, u_2\}$. Furthermore, since there is no type-1 edge, C is induced and hence a basic 4-cycle, a contradiction.

If $e = v_1v_2$ is a type-3 edge, then again there is a vertex y_i both inside and outside the cycle $C : x, u_1, v_1, v_2, u_2, x$ not dominated by $\{x, u_1, u_2\}$. Furthermore, since there is no type-1 or type-2 edge, C is induced and hence a basic 5-cycle, a contradiction.

(ii) If there is an edge from Y_i to Y_j where i and j are not consecutive, then the edge is type-3, contradicting (i) above. □

For an index i, if $y_i \sim y_{i+1}$ we define the lozenge L_i as the region inside the 5-cycle $x, n_i, y_i, y_{i+1}, n_{i+1}, x$.

Claim 4.1.2 *(i) If there is a path of length* 2 *from* $y \in Y_i$ *to* $z \in Y_{i+2}$*, then the intermediate vertex is* y_{i+1}*.*
(ii) The shortest y_i*–*y_{i+3} *path is* $y_i, y_{i+1}, y_{i+2}, y_{i+3}$*, or if* $m = 6$ *possibly* $y_i, y_{i-1}, y_{i-2}, y_{i+3}$*.*
(iii) If y *is a vertex of* Y *inside lozenge* L_i*, then the shortest* y*–*y_{i+3} *path is* $y, y_{i+1}, y_{i+2}, y_{i+3}$*. In particular,* y *is adjacent to both* y_i *and* y_{i+1}*.*

Proof. (i) Assume the path is y, a, z. Then $a \notin N(x)$, else one of the edges ay and az would be a type-2 edge. So $a \in Y$ and in fact $a \in Y_{i+1}$ by part (ii) of Claim 4.1.1. Further, vertex a cannot be placed in Y_i or Y_{i+2}, that is, $a \in Y'_{i+1}$; and, by the definition of y_{i+1}, in fact $a = y_{i+1}$.

(ii) Assume the shortest y_i–y_{i+3} path P is y_i, a, b, y_{i+3}. Suppose both a and b in $N(x)$. Since P has no type-2 edge, a lies between n_{i-1} and n_{i+1} while b lies between n_{i+2} and n_{i+4}, and so the middle edge of P is a type-1 edge, a contradiction. Suppose exactly one of a and b is in Y, say a. By part (ii) of Claim 4.1.1, a is in Y_i, Y_{i-1} or Y_{i+1}. But then the second or third edge of P is a type-2 edge, again a contradiction.

Hence both a and b are in Y. Thus, $a \in Y_{i+1}$ and $b \in Y_{i+2}$, or possibly $a \in Y_{i-1}$ and $b \in Y_{i-2}$ if $m = 6$. In fact, by part (i) of this claim, the path P is either $y_i, y_{i+1}, y_{i+2}, y_{i+3}$ or, if $m = 6$, possibly $y_i, y_{i-1}, y_{i-2}, y_{i+3}$.

(iii) Assume the shortest y–y_{i+3} path is y, a, b, y_{i+3}. Then $a \notin \{n_i, n_{i+1}\}$, since each of n_i and n_{i+1} is distance at least 3 from y_{i+3} (if for example n_{i+1} and y_{i+3} have a common neighbour then there is a short basic cycle). So $a = y_{i+1}$ and by part (i), $b = y_{i+2}$.

Similarly, the shortest y–y_{i-2} path is y, y_i, y_{i-1}, y_{i-2}. In particular, $y \sim y_i$ and $y \sim y_{i+1}$. □

By Claim 4.1.2, we may assume that $m \leq 7$, for otherwise $d(y_0, y_4) > 3$. So there are two cases.

Case 1: $m = 7$.

Then a shortest path from y_i to y_{i+3} must have as intermediate vertices y_{i+1} and y_{i+2}. That is, there is an edge from y_i to y_{i+1} for all i. Suppose there is another vertex of Y; say inside the lozenge L_0. Then it is too far from y_4, a contradiction. Hence, $Y = \{y_0, y_1, \ldots, y_6\}$ and thus $\{x, y_0, y_2, y_4\}$ dominates G.

Case 2: $m = 6$.

Consider the shortest y_0–y_3, y_1–y_4 and y_2–y_5 paths. Since by Claim 4.1.2(ii) each interior vertex on any such path is a vertex y_i for some i, it follows that there are at least five consecutive edges $y_i y_{i+1}$, starting at y_0 say. By Claim 4.1.2(iii), any vertex of Y inside $L_0 \cup L_1$ is adjacent to y_1, and any vertex of Y inside $L_2 \cup L_3$ is adjacent to y_3. Hence $\{x, n_0, y_1, y_3, n_4, n_5\}$ dominates G. Consequently, $\gamma(G) \leq 6$. $\Box$

4.2 Cut-Cycles and Domination

To prove Theorem 4, we need a series of lemmas, the key lemma of which is Lemma 4.4. We define the *distance* of a vertex v from a set S as the minimum distance from v to a vertex of S.

Lemma 4.2 *Let $C : x, y, z, x$ be a triangle in a plane graph G and let S be the set of vertices inside C that are at distance exactly 2 from each vertex of C. Then S is dominated by two vertices.*

Proof. Let s be a vertex of S; i.e., $d(s, x) = d(s, y) = d(s, z) = 2$. If a neighbour of s is adjacent to each of x, y and z, then that vertex dominates S since it separates each vertex of S from one of x, y or z. If each neighbour of s is adjacent to at most one of x, y and z, then s is the unique vertex of S by the planarity of G. So we may assume a neighbour of s, say v, is adjacent to two vertices of C, say x and y. Let w be a common neighbour of z and s. Then $\{v, w\}$ dominates S by the planarity of G. $\Box$

Lemma 4.3 *Let $C : w, x, y, z, w$ be a 4-cycle in a plane graph G and let S be the set of vertices inside C that are at distance exactly 2 from each vertex of C. Then S is dominated by one vertex.*

Proof. Let s be a vertex of S; i.e., $d(s, w) = d(s, x) = d(s, y) = d(s, z) = 2$. If a neighbour u of s is adjacent to opposite vertices of C, say to both x and z, then the vertex u dominates S since the path x, u, z separates each vertex of S from one of w or y. So we may assume that no neighbour of a vertex in S is adjacent to opposite vertices of C. If each neighbour of s is adjacent to at most one vertex in C, then s is the unique vertex of S by the planarity of G. So we may assume some neighbour of s, say a, is adjacent to two consecutive vertices of C, say w and x.

Since neither ay nor az are edges of G, there can be no vertex of S inside the region w, a, x, w. If another neighbour of s is adjacent to both y and z, then for any vertex of S the shortest path either to w or to x must be via a; so the vertex a dominates S.

On the other hand, suppose c is a common neighbour of s and y, and d is a common neighbour of s and z where $c \neq d$. By the planarity of G, at most one of cz and dy is an edge of G; we may assume $d \not\sim y$. Thus there is no vertex of S inside z, w, a, s, d, z, for otherwise such a vertex would be at distance at least 3 from y. In fact, any shortest path from a vertex of S to one of x or z must be via the vertex c. Hence the vertex c dominates S. □

Lemma 4.4 *Consider a planar graph G. Let x and y be distinct vertices such that $G + xy$ is planar. Let S be the set of vertices at distance exactly 2 from both x and y. Suppose every pair of vertices in S are distance at most 3 apart. Then S is dominated by three vertices.*

Proof. Embeddibility on the sphere and the plane is identical. Therefore, we may draw $G + xy$ (or G if the edge xy already exists) in the plane with x directly above y, and with the edge xy extending from x to the top of the page, wrapping around the back of the page, and then extending from the bottom of the page to y. The result is that each x–y path other than the edge xy divides the plane into a left and right part.

For each vertex s in S, we define the *s-connectors* as follows. If $U_s = N(s) \cap N(x) \cap N(y)$ is nonempty, then every x–y path of length 2 with the internal vertex in U_s is an s-connector. On the other hand, if $U_s = \emptyset$, then every path of length 4 consisting of x–s and s–y shortest paths is an s-connector. We define a *connector* to be an s-connector for some $s \in S$. Note that a length-4 connector does not contain a vertex of $N(x) \cap N(y)$.

We say that a connector P is *to the left* of a connector Q if all the edges of P are either in Q or to the left of Q. Similarly we say that P is to the right of Q if all the edges of P are either in Q or to the right of Q. These two notions induce two partial orderings on the set of connectors.

Claim 4.4.1 *We can choose a leftmost connector and a rightmost connector.*

Proof. Suppose there are two different connectors P and Q which do not have connectors to the left. Then there are edges of Q to the left of and to the right of P. We show that one can find a connector R to the left of both P and Q as follows. If P has length 2, then Q contains a vertex of $N(x) \cap N(y)$ and hence has length 2 too. So $P = Q$, a contradiction. Hence we may assume that both P and Q have length 4. If P is an s-connector and Q is an s'-connector, where $s \neq s'$, then P and Q have a common internal vertex which is adjacent to x or y. But then there can be no subsection of Q that is both to the left of and to the right of P, a contradiction. Hence P and Q are both s-connectors. The leftmost s–x and the leftmost s–y subpaths of P and Q form the desired connector R. □

Let u be the central vertex of the leftmost connector P. If P has length 4, then there can be no vertex of S to the left of P and we let $s = u$. On the other hand, if P has length 2, then either there is vertex of S to the left of P, in which case we choose one such vertex and call it s (necessarily s is adjacent to u), or every vertex of S is to the right of P, in which case we let s be any vertex of $S \cap N(u)$. Similarly, let v be the central vertex of the rightmost connector Q. If Q has length 4, then we let $t = v$. If Q has length 2, then either there is vertex of S to the right of Q, in which case we choose one such vertex and call it t, or every vertex of S is to the left of Q, in which case we let t be any vertex of $S \cap N(v)$.

By examining the possibilities, one can easily show that if P and Q are not internally disjoint, then any common vertex dominates S. So we may assume they are internally disjoint. Let R be the region inside the cycle determined by P and Q. Let S^* be the set of vertices at distance at least 2 from both x and y.

Claim 4.4.2 *If W is any u–v walk containing neither x nor y, then $W - S^*$ dominates S except possibly for vertices of S on W.*

Proof. If there is a vertex of S to the left of P, then it is adjacent to u, and $u \notin S^*$. If there is a vertex of S to the right of Q, then it is adjacent to v, and $v \notin S^*$. If there is a vertex of S on P, then it is u and hence in W. If there is a vertex of S on Q, then it is v and hence in W. So we need only consider vertices r of S inside R but not on W.

The portion of W from the last vertex on P to the first vertex on Q separates R into two regions, say R_x and R_y with x and y, respectively, on the boundary. Without loss of generality, r lies inside R_x. Then r and y have a common neighbour r^*. This common neighbour cannot be on $P - W$ or $Q - W$, since then r^* would be in $N(x) \cap N(y)$, and would contradict the definition of P or Q being a connector. So r^* is on W. Clearly $r^* \notin S^*$. So $W - S^*$ dominates r. □

Now, by assumption, there is an s–t path of length at most 3. Let M be a shortest s–t path. Let W be the u–v walk obtained by starting at u, proceeding along the edge from u to s if $u \neq s$, following the path M from s to t, and then proceeding along the edge from t to v if $t \neq v$. (Note that if s outside R then the first two edges of W are the same.)

There are two cases to consider.

Case 1: At least one connector is long.

If M has length 1 then W dominates S; otherwise $W - \{s, t\}$ dominates S. In either case S is dominated by at most 3 vertices.

Case 2: Both connectors are short.

If M has length 1 or 2, then $W - \{s, t\}$ dominates S. So we may assume M has length 3. If either interior vertex of M is in S^*, then u, v and the other interior vertex dominates S, by above claim. In fact, we are done unless:

> *for every vertex $s \in S \cap N(u)$ and every $t \in S \cap N(v)$, every shortest s–t path has length 3 with neither internal vertex in S^*.*

Now, choose s and t closest to y as given by the ordering of the vertices at u and v in the embedding of G (or $G + xy$). Assume the s–t path is s, m, n, t. If $\{u, m, n\}$ or $\{m, n, v\}$ dominates S then we are done. So there exists a vertex $s' \in S \cap N(u)$ not dominated by $\{v, m, n\}$ and a vertex $t' \in S \cap N(v)$ not dominated by $\{u, m, n\}$. Choose s' and t' closest to y as given by the ordering of the vertices at u and v in the embedding of G.

Now, a shortest s'–t' path is disjoint from M (by the definition of s' and t'), and so it is s', m', n', t' say. By Claim 4.4.2, $\{u, m', n', v\}$ dominates S. We know that $m, n, m', n' \notin S^*$. So $m \sim y$, $n \sim y$, $m' \sim x$, $n' \sim x$. See Figure 9.

Now, assume $\{u, m', n'\}$ does not dominate some vertex t'' of S. By Claim 4.4.2, $t'' \sim v$. If t'' lies above t' (i.e., inside the region x, n', t', v, x), then the length-3 path from t'' to s

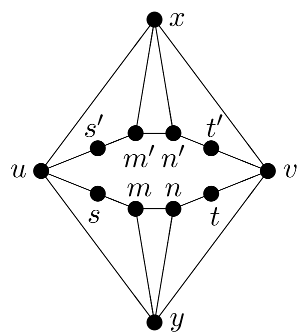

Figure 9: A subgraph of G

must use the vertex v. But this creates a u–v path W' with length at most 3, and which contains s. So $W' - \{s\}$ has three vertices and dominates S, and we are done. Hence by the choice of t, we may assume t'' is t or lies between t and t'. If t'' lies between t and t', then by our choice of t', it follows that t'' is dominated by $\{u, m, n\}$. So by planarity, $n' \not\sim t$. On the other hand, if $t'' = t$ then $n' \not\sim t$. It follows that $n' \not\sim t$. By a similar argument, $s \not\sim m'$.

Finally, consider a s–t' shortest path. This has length 3 and each intermediate vertex is not in S^*. Thus this path must be s, m, n', t'. Similarly, a s'–t shortest path must be s', m', n, t. But we cannot have both edges mn' and $m'n$ by planarity, a contradiction.

So, there is a set of three vertices that dominates S. □

In using Lemma 4.4, we can start with disjoint sets X and Y which each induce a connected subgraph and then contract the vertices in X to a single vertex x and the vertices in Y to a single vertex y without affecting the number of vertices needed to dominate S. Hence we have the following consequence of Lemma 4.4.

Corollary 4.5 *Let X and Y be disjoint sets in a plane graph G that both induce connected subgraphs and let S be the set of vertices at distance exactly 2 from both X and Y. Suppose there is an edge from X to Y and any pair of vertices of S are distance at most 3 apart. Then S is dominated by three vertices.*

4.3 Basic Cycles and Domination

In this section, we use Lemma 4.4 to bound the domination number from above when there exist a short basic cycle.

Lemma 4.6 *Let G be a plane graph of radius 2 and diameter 3 with central vertex x. If there exists a basic triangle C, then $\gamma(G) \leq 6$.*

Proof. Let C be the triangle x, a, b, x and let S be the set of vertices of G not dominated by C. Since G has diameter 3, the vertices in S are confined to one side of C, say the inside of C. Since C is basic, there is a vertex outside C nonadjacent to x. Hence every vertex of S is within distance 2 from $\{a, b\}$. Let $X = \{x\}$ and $Y = \{a, b\}$; since G has diameter 3, by Corollary 4.5, the set S is dominated by three vertices. Consequently, $\gamma(G) \leq |V(C)| + 3 = 6$. $\square$

Lemma 4.7 *Let G be a plane graph of radius 2 and diameter 3 with central vertex x. If there exists a basic 4-cycle C, then $\gamma(G) \leq 6$.*

Proof. Let C be the 4-cycle x, a, b, c, x. Let S_C be the set of vertices not dominated by C. Since G has diameter 3, the vertices in S_C are confined to one side of C, say the inside of C. Since C is basic, there is a vertex outside C nonadjacent to both x and (say) a. Hence every vertex of S_C is within distance 2 from $\{b, c\}$. In fact, since any neighbour of a is within distance 2 of $\{b, c\}$, every vertex of G is within distance 2 from $\{b, c\}$. Let S_a be the set of neighbours of a at distance 2 from $\{x, b, c\}$.

Let $X = \{x\}$, $Y = \{b, c\}$, and $S = S_C \cup S_a$. Since G has diameter 3, by Corollary 4.5, the set S is dominated by three vertices. The vertices of G not in S are dominated by $\{x, b, c\}$. Consequently, $\gamma(G) \leq 6$. $\square$

Lemma 4.8 *Let G be a plane graph of radius 2 and diameter 3 with central vertex x. If there exists a basic 5-cycle C, then $\gamma(G) \leq 6$.*

Proof. Let C be the 5-cycle x, a, b, c, d, x and let E be the set of vertices not dominated by C. Since G has diameter 3, the vertices in E are confined to one side of C, say the inside of C. Since C is basic, there is a vertex f outside C nonadjacent to all of x, a and d. Hence every vertex of E is within distance 2 from $\{b, c\}$ in order to reach f. In fact, since any neighbour of a or d is within distance 2 of $\{b, c\}$, every vertex of $G - N(x)$ is within distance 2 from $\{b, c\}$.

Let H be obtained from the plane subgraph induced by the vertices on and inside the cycle C by adding an artificial edge xc outside C. Let S_H be the set of vertices of H at distance 2 from $\{x, b, c\}$. Then $E \subseteq S_H$. Since G has diameter 3, for any $s_1, s_2 \in S_H$, there exists an s_1–s_2 path of length at most 3 in G. However, since $a \not\sim d$, such a path remains on or inside C. Let $X = \{x\}$ and $Y = \{b, c\}$. Corollary 4.5, applied to H, implies that the set S_H is dominated by three vertices. This shows that if every vertex outside C is dominated by $\{x, b, c\}$, then $\gamma(G) \leq 6$.

So, we may assume that there is a vertex outside C nonadjacent to all of x, b and c. Since f is adjacent to one of b or c, and there is an x–f path of length 2 (necessarily via a vertex outside C), a nonneighbour of x outside C cannot be adjacent to both a and d. Hence, any vertex outside C not dominated by $\{x, b, c\}$ is adjacent either only to a or only to d.

Suppose there is outside C both a vertex a' adjacent only to a on C and a vertex d' adjacent only to d on C. Then every vertex of E is distance 2 from a and distance 2 from d. Hence letting F be the plane subgraph obtained from G by shrinking $\{b, c\}$ to a single vertex y, each vertex in E is at distance exactly 2 from each vertex in $\{x, a, y, d\}$. By Lemma 4.3 applied to F, it follows that E can be dominated by one vertex. Adding that vertex to $V(C)$ produces a dominating set of G of cardinality 6.

On the other hand, suppose every vertex outside C that is not dominated by $\{x, b, c\}$ is adjacent only to a on C. By assumption there is at least one such vertex outside C. Then every vertex of E is at distance 2 from a, and $\{x, a, b, c\}$ dominates the vertices outside C. Let H^* be the plane subgraph obtained from H by shrinking $\{b, c\}$ to a single vertex y. Let S^* be the set of vertices of H^* at distance 2 from the triangle x, a, y, x. Then each vertex of S^* is at distance exactly 2 from every vertex of the triangle. So we may apply Lemma 4.2 to show that S^* can be dominated by two vertices. Adding these two vertices to the set $\{x, a, b, c\}$ forms a dominating set of G of cardinality 6. □

Theorem 4 now follows immediately from Lemmas 4.1, 4.6, 4.7 and 4.8.

5 Proof of Theorem 5

We recall a family of graphs known as *lanterns* (or *theta graphs*). For $s \geq 3$, an *s-lantern* is a graph obtained from the complete bipartite graph $K(2, s)$ by subdividing each edge any number of times (including the possibility of none). The two vertices of degree more than 2, say x and y, we call the *hubs* of the lantern and the x–y paths of the lantern we call the *axes* of the lantern. (A 5-lantern with hubs x and y is illustrated in Figure 10.) A lantern with hubs x and y we also call an *x-y lantern*. A *region* of a lantern is a portion of the plane bounded by two consecutive axes in the lantern.

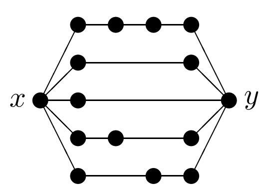

Figure 10: A 5-lantern with hubs x and y

The following lemma establishes for any $s \geq 3$ the existence of an s-lantern in a graph of sufficiently large order.

Lemma 5.1 *Let d be a positive integer. In a sufficently large graph of diameter d and radius d there exists an arbitrarily large lantern.*

Proof. Consider a graph of order n with diameter and radius d. Then the maximum degree is at least $(n-1)^{1/d}$. Now, let x be a vertex of maximum degree, and let w be a vertex at distance d from x. The distance from w to each of x's neighbours is at most d and hence a shortest path from w to a neighbour of x does not use x.

Now, consider a tree T formed by a collection of shortest paths from w to each neighbour of x. If the degree of x is sufficiently large, then T contains a vertex of large degree, call it y. By using the edges of T and the edges incident with x, we can obtain a large lantern with x and y as hubs. □

Lemma 5.2 *Let G be a planar graph of radius 3 and diameter 3. If G contains a 10-lantern L, then there is an axis of L whose contraction to a single vertex produces a planar graph G' of radius at most 2 and diameter at most 3 whose domination number is at least $\gamma(G) - 1$.*

Proof. Let x and y be the hubs of L. Let S be the vertices at distance at least two in G from $\{x, y\}$ and let v be any vertex of S. Then since G has diameter 3, the four consecutive regions and three axes of L fartherest from v cannot contain a vertex of S. In fact, S is confined to four consecutive axes and the three regions between them. From this it follows that there is a nontrivial axis B such that every vertex of S is distance 3 from internal vertices of B. Also, for v to be within distance 3 from every vertex of the lantern L, it follows that the distance from v to the set $\{x, y\}$ must be 2.

Now, let A be any axis different from B not containing a vertex of S. Let G' be the graph obtained by contracting A (including the vertices x and y) to a single vertex z. Clearly G' is planar with diameter at most 3. Moreover, every vertex is within distance 2 from z in G'. Thus, G' has radius at most 2.

Now, consider a minimum dominating set D of G'. Let w be a vertex of D that dominates a vertex of B. Let $D' = (D - \{w\}) \cup \{z\}$. Suppose some vertex s of G' is not dominated by D'. Then $s \in S$ and s is within distance 1 of w. However, w is at distance at most 1 from B so that s is at distance at most 2 from B, a contradiction. Hence D' is also a dominating set of G'.

Thus $(D' - \{z\}) \cup \{x, y\}$ is a dominating set of G. Consequently, $\gamma(G) \leq |D'| + 1 = \gamma(G') + 1$, as required. □

A consequence of the above two lemmas is:

Lemma 5.3 *For a sufficiently large planar graph G of radius and diameter 3, there exists a planar graph G' of radius at most 2 and diameter at most 3 such that $\gamma(G) \leq \gamma(G') + 1$.*

The requirement for sufficiently large in the above lemma as given by the proofs of the previous two lemmas is rather large and can certainly be reduced with some more work. It is possible that in fact the requirement is rather small, but a proof of this seems to require a different approach.

Theorem 5 now follows immediately from Theorem 4 and Lemma 5.3.

6 Other Surfaces

Theorem 6 *For each surface, there are finitely many graphs with diameter 2 and domination number more than 2.*

Proof. Consider a given surface, and let m be the maximum integer such that $K(3, m)$ embeds on the surface (see [1]). Let G be a sufficiently large graph on that surface. By Lemma 5.1, G has either a dominating vertex or a large lantern. We may assume the latter. In fact, we may assume that G has an $(m^2 + m + 1)$-lantern L.

Say the hubs of the lantern L are x and y and let $S = V(L) - \{x, y\}$. We claim that $\{x, y\}$ dominates G. Suppose there is a vertex z which the pair $\{x, y\}$ does not dominate.

Note that each axis of L has at most two interior vertices. Then consider the subtree T consisting of a shortest path from z to each axis. (Note that z is a cental vertex and the tree has radius at most 2.) There are two cases.

Suppose first that there is a vertex z' of T adjacent in T to $m + 1$ end-vertices of T; call this set of end-vertices S'. Then there is a subdivision of $K(3, m + 1)$ with one partite set S' and the other $\{x, y, z'\}$. (Note that the vertices of S' are on distinct axes.) So we contradict the value of m.

On the other hand, suppose there is no vertex of T adjacent in T to $m + 1$ end-vertices of T. Then since there are more than $m^2 + m$ axes, in T vertex z has $m + 1$ non-end-vertex neighbours, call the set N'. For each vertex of N', pick an end-vertex of T adjacent to it in T; call the resultant set S'. This yields a subdivision of $K(3, m + 1)$ with one partite set S' and the other partite set $\{x, y, z\}$. (Note that a vertex of N' and a vertex of S' cannot be on the same axis by the way T is constructed.) So we contradict the value of m.

That is, the set $\{x, y\}$ dominates G. $\Box$

Theorem 7 *For each orientable surface, there is a maximum domination number of graphs with diameter* 3.

We use a crude argument with no attempt to provide even a reasonable bound for the value. We need two lemmas, the first of which is probably known.

Lemma 6.1 *For any nonnegative integers g and d, there exists a constant $c_{g,d}$ such that for every graph G of genus at most g and diameter at most d there is a set S of at most $c_{g,d}$ vertices such that $G - S$ is planar.*

Proof. Assume G has genus g and diameter d. Then create a series of graphs $G = G_g, G_{g-1}, \ldots, G_0$ as follows. Let C_i be a minimum noncontractible cycle in G_i. Form G_{i-1} from G_i by splicing C_i down the middle; in this process the cycle C_i is replaced by two copies C_i^1 and C_i^2, and each edge with exactly one end in C_i is replaced by one edge with exactly one end in $C_i^1 \cup C_i^2$. We call the vertices of $\bigcup_i (C_i^1 \cup C_i^2)$ *split* vertices. The graph G_i has genus i (see [1]).

Let S_i denote the set of all split vertices in G_i; $|S_i| = s_i$. We shall now establish an upper bound on s_i. A vertex which is not a split vertex we call an *original* vertex. Let B_i be a collection of original vertices pairwise at least distance $2d + 1$ apart on C_i. Since C_i is a minimum noncontracible cycle of G_i, the vertices of B_i are pairwise at least distance $2d + 1$ apart in G_i. Suppose that $|B_i| \geq s_i + 2$.

Pick any vertex b^* in B_i. For every other vertex b of B_i, consider a shortest path P_b from it to b^* in the original graph G. Define a vertex $w_b \in V(G_i)$ as follows. If all the vertices of P_b are in G_i, then let $w_b = b^*$. Otherwise, let v be the first vertex of P_b that no longer exists, and u the vertex on P_b before it. Then let w_b be the split vertex derived from v which is adjacent to u in G_i. There is a b–w_b path in G_i of length at most d.

Since $|B_i - \{b^*\}| > s_i$, by the pigeon-hole principle there is a pair of vertices b, b' in $B_i - \{b^*\}$ such that $w_b = w_{b'}$. Hence b and b' are at distance at most $2d$ apart in G_i, a contradiction. It follows that $|B_i| \leq s_i + 1$.

Since we can choose B_i greedily, skipping S_i, it follows that $|V(C_i)| \leq (2d+1)|B_i| + s_i < (2d+2)(s_i+1)$. Hence $s_{i-1} \leq 2|V(C_i)| + s_i < (4d+5)(s_i+1)$.

Finally, let S be the set of all vertices of G which do not appear in G_0. Then $|S|$ is bounded above by s_0 and $G - S$ is a subgraph of G_0, which is planar. $\square$

The next lemma generalises the technique of the proof of Lemma 4.4.

Lemma 6.2 *For any planar graph H and any subset T of the vertices such that every two vertices in T are distance at most 3 apart, T can be dominated by at most 38 vertices.*

Proof. For a vertex x, let e_x denote the maximum distance from x to a vertex of T.

Claim 6.2.1 *Under the hypotheses of the lemma, let vertices x and y of T be such that $\max(e_x, e_y) \leq 2$, or x and y are nonadjacent, or both. Then there is a set B, containing neither x nor y, $|B| \leq 12$, such that every vertex of $T - N[x] - N[y]$ is within distance $\max(e_x, e_y) - 1$ of B and contracting B to a single vertex preserves planarity.*

Proof. Let $T_{xy} = T - N[x] - N[y]$. There are two cases, depending on $\kappa(x, y)$, the maximum number of internally disjoint paths between x and y. If $\kappa(x, y) \leq 9$, then there exists a minimal set B of vertices, $|B| \leq 9$, that separates x and y. Every vertex of T is within distance $\max(e_x, e_y) - 1$ of B (as B separates it from at least one of x or y). Since B is a minimal separating set in a planar graph, it can be contracted to a single vertex while preserving planarity.

If $\kappa(x, y) \geq 10$, then there are at least 10 internally disjoint paths from x to y. So there exists one of these paths, call it D, that is not used for any shortest t_1–t_2 path, $t_1, t_2 \in T_{xy}$, nor for any shortest t_1–x or t_1–y path (since vertices in T are distance at most 3 apart).

Following the proof of Lemma 4.4, for each vertex $s \in T_{xy}$ define a connector as any combination of a shortest x–s path and a shortest s–y path. (Note that by the limitation on x and y, the shortest s–x path cannot contain y and vice versa.) Then by wrapping the divider D around the back of the page, every x–y path internally disjoint from D divides the plane into a left and right part. So we can find a leftmost connector P (in that every other connector is either to the right of P or crosses P); say it is a connector for s. Similarly we can find a rightmost connector Q, say for t. (Note that there might be several leftmost and rightmost connectors.)

Now let W be a shortest s–t path (it does not use D or $\{x, y\}$). Let B be the set of vertices of P, Q and W except for x, y. Then every vertex u of T_{xy} is within distance $\max(e_x, e_y) - 1$ of B. (If u lies between P and Q then the path W separates it from one of x or y; if u lies outside P and Q, and the shortest u–x and u–y paths are both disjoint from B, then that contradicts the choice of P or Q as outermost connector.)

Since $e_x \leq 3$ and $e_y \leq 3$, the x–s subpath of P, the s–y subpath of P, the x–t subpath of Q, and the t–y subpath of Q each contain at most two internal vertices. Furthermore, the path W contains at most four vertices (including s and t). It follows that $|B| \leq 12$. Since B induces a connected subgraph, it can be contracted to a single vertex while preserving planarity. $\square$

We now return to the proof of Lemma 6.2. We start with any nonadjacent vertices x and y of T. By hypothesis, $e_x, e_y \leq 3$. Then we apply the above claim to obtain a set B of vertices which when shrunk to a single vertex b has $e_b \leq 2$. If $e_b = 1$ then $B \cup \{x, y\}$ dominates T, so we may assume $e_b = 2$. Then let z be a vertex of T nonadjacent to b. By the above claim, with vertices b and z, we obtain a set B' of vertices which when shrunk to a single vertex b' has $e_{b'} \leq 2$.

Finally, by the claim with vertices b and b', we obtain a set B'' such that every vertex of T not dominated by $\{x, y, b, b'\}$ is within distance $\max(e_b, e_{b'}) - 1 = 1$ of B''. Hence $\{x, y\} \cup B \cup B' \cup B''$ dominates T in H, and the lemma is true. $\square$

To prove the theorem, apply Lemma 6.1: let $H = G - S$ be the resulting planar graph and T the set of vertices of G not dominated by S. The vertices of T are pairwise at most distance 3 apart in H as there was a path of length 3 in G and this path is still there in H. Then apply Lemma 6.2 to H to obtain set B to dominate T. The set $S \cup B$ dominates G and is bounded.

7 Open Questions

The most obvious open question is what is the maximum domination number of a planar graph of diameter 3. We have shown that there are finitely many planar graphs of diameter 3 with domination number more than 7. It is entirely possible that there is none.

It is easy to show that this maximum is at most 4 more than the maximum for radius 2 and diamter 3. For example, if G has diameter and radius 3, then shrink the open neighbourhood of a vertex v of minimum degree to a single vertex x. The resultant graph G' has diameter at most 3 and radius 2. Also, $\gamma(G') \leq \gamma(G) + 4$ since we may assume x is in the minimum dominating set S' of G', and $(S' - \{x\}) \cup N(v)$ dominates G. (Also, with Theorem 4 this observation recovers Theorem 3.)

We believe that Theorem 4 can be improved to an upper bound of 5. This would improve Theorem 5 to an upper bound of 6, both of which are best possible. It would also improve on Theorem 3 to show that a planar graph of diameter 3 has domination number at most 9, by the observation in the previous paragraph. To this end, we have made good progress. However, there remains a significant gap that we have yet to settle. More precisely, we define a *special* basic cycle as one with the added or stronger condition that there is on the dominated side an exterior vertex with only one neighbour on the cycle and that neighbour is not x. With considerable work, we can show that if G is a plane graph of radius 2 and diameter 3 that contains no special basic cycles of length 4 or 5, then $\gamma(G) \leq 5$.

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